

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Computer Vision with OpenCV COURSE CODE: DSA02

COURSE OUTCOME COVERED

CO3: Analyze shape representation methods and techniques such as contours, regions, deformable models, and multi-resolution analysis. (BL4)

Assessment Tool Weightage: 30%

QUESTIONS: 2 Total Marks:30 Annexure A

Case study

Code of Conduct:

I Cheredla Jasvina / 192324026S certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	

Clarity	10	Clear, well structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	
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Annexure A

Case study

Case Study 1: Defect Detection in Industrial Parts

A manufacturing company wants to automatically detect defects in gears and machine components using computer vision. The images of the components are binary and may contain small protrusions or missing parts.

Questions:

A. How can region-based representation and medial axis computation help in identifying defective parts?

Answer :

Region-based representation considers the entire binary region of the industrial part rather than only its boundary. The binary image is segmented into foreground (part) and background.

It can help detect defects by measuring:

- **Area:** A missing part decreases the area.
- **Perimeter:** A protrusion or irregularity increases the perimeter.
- **Centroid:** A defect may shift the center of the component.
- **Shape and compactness:** Changes indicate abnormal geometry.
- **Connected components:** Helps identify separated or missing regions.

Medial axis (skeleton) reduces the object to a thin centerline while preserving its basic **shape and connectivity**. **For gears and machine components, it can be used to:**

- Compare the skeleton of a test component with a normal component.
- Detect missing teeth or protrusions as changes in skeleton branches.
- Identify changes in thickness and structural symmetry.
- Detect broken or disconnected regions.

B. Discuss how noise in image acquisition could affect shape analysis and propose preprocessing steps to improve accuracy.

Answer :

Noise during image acquisition can create unwanted pixels around the component. This can seriously affect shape analysis.

For example:

- Salt-and-pepper noise can create small unwanted protrusions.
- Missing pixels can make a normal part appear defective.
- Noise can change area and perimeter.
- It can create false branches in the medial axis.
- Boundary noise can produce incorrect Fourier or wavelet descriptors.

Preprocessing steps

1. **Noise filtering:** Apply a median filter to remove salt-and-pepper noise.
2. **Gaussian filtering:** Reduce high-frequency noise and smooth the image.
3. **Thresholding:** Convert the image into a clean binary image.

4. **Morphological opening:** Remove small unwanted objects and protrusions.
5. **Morphological closing:** Fill small gaps and holes.
6. **Connected-component analysis:** Remove isolated noise regions.
7. **Skeleton pruning:** Remove small unwanted branches after medial-axis computation.

These steps produce a cleaner shape and improve the reliability of defect detection.

C. Compare the effectiveness of Fourier descriptors versus wavelet descriptors for classifying normal vs. defective components.

Answer :

Fourier Descriptors:

- Represent the shape using frequency components of the boundary.
- Good for describing the overall shape of gears and machine parts.
- Effective for detecting large shape differences.
- Less effective for very small and localized defects.

Wavelet Descriptors:

- Represent the shape at different scales and locations.
- Can detect both overall shape changes and small local defects.
- More effective for detecting small protrusions, missing teeth, cracks, and irregularities.
- Provides better information about where the defect occurs.

Comparison:

Fourier descriptors are suitable for global shape classification, while wavelet descriptors are more effective for local defect detection. Therefore, for classifying normal and defective industrial components with small defects, wavelet descriptors are generally more effective.

Case Study 2: Medical Image Segmentation

A hospital is developing a system to segment organs from CT scans. The organ boundaries are irregular and may vary slightly between patients. Questions:

A. Explain how active contours (snakes) can be applied to segment the organ boundaries.

Answer :

Active contours, also called snakes, are deformable curves used to identify the boundary of an organ in a CT image. The contour starts from an initial position and gradually moves toward the actual boundary of the organ.

The process works as follows:

1. Preprocess the CT image:

Noise is reduced and the image may be enhanced to make the organ boundary clearer.

2. Initialize the contour:

A closed curve is placed around or near the organ to be segmented.

3. Contour deformation:

The contour is allowed to move and change its shape. It is controlled by internal and external forces.

4. Internal forces:

These forces control the smoothness and continuity of the contour. They prevent the contour from becoming unnecessarily irregular.

5. External forces:

These forces come from the image. Strong edges in the CT image attract the contour toward the organ boundary.

6. Energy minimization:

The snake moves until its total energy becomes minimum. At this point, the contour generally fits the organ boundary.

7. Final segmentation:

The region inside the final contour is considered the segmented organ.

Example:

If a CT image contains a liver, an initial contour can be placed around the liver. The snake gradually moves toward the liver's boundary and adapts to its irregular shape. Finally, the area enclosed by the contour represents the segmented liver.

B. Discuss the challenges of initializing contours and tuning parameters for accurate segmentation.

Answer :

Accurate active-contour segmentation depends heavily on where the contour is initialized and how its parameters are selected.

1. Initialization of contours

- The initial contour is the starting curve of the snake.
- If the contour is too far from the organ, it may move toward another strong edge instead of the desired organ.
- If the contour is inside the organ, it may fail to reach the complete boundary.
- If the organ has a complex or irregular shape, choosing a suitable initial contour becomes difficult.
- Different patients may have different organ sizes and shapes, so the same initial contour may not work for every CT scan.

Example:

If the liver boundary is irregular and the initial contour is placed incorrectly, the snake may stop at nearby blood vessels or other strong edges instead of the liver boundary.

2. Parameter tuning

Active contours have several parameters that control their behavior.

Elasticity: Controls how much the contour stretches.

Smoothness: Controls how smoothly the contour bends.

Image/edge force: Determines how strongly the contour is attracted to image boundaries.

Step size: Controls how quickly the contour moves.

Stopping condition: Determines when the contour should stop moving.

If the parameters are not properly selected:

- The contour may become too smooth and miss important boundary details.
- The contour may become too flexible and follow noise.
- Weak organ boundaries may not attract the contour sufficiently.
- The contour may stop at an incorrect boundary.
- Segmentation may be incomplete

C. How could level set methods improve segmentation for cases where organs split or merge in the images?

Answer :

Level set methods are an advanced technique for image segmentation in which the boundary is represented using a mathematical function called a level set function.

Unlike a traditional active contour, a level set does not require the boundary to maintain a fixed shape or topology. This makes it useful for medical images where structures can split, merge, or change shape.

How it works:

1. An initial contour or level set is defined around the organ.
2. The level set evolves according to information from the image.
3. The boundary moves toward the desired organ boundary.
4. If one region needs to split into two regions, the level set can automatically split.
5. If two regions need to join together, the level set can automatically merge.
6. The final level set represents the segmented organ or organs.

Example of splitting:

Suppose two separate parts of an organ appear as disconnected regions in a CT image. A level set method can automatically divide the evolving contour into two separate boundaries.

Example of merging:

If two portions of the same organ appear connected in the image, the evolving level set can merge their boundaries into a single region.

Advantages of Level Set Methods

- Can handle complex and irregular shapes.
- Automatically handles splitting and merging of regions.
- Does not require manually specifying the final shape.
- Can handle topology changes naturally.
- More flexible for medical images containing complicated structures.

Conclusion

Active contours are effective for following irregular organ boundaries, but their performance depends strongly on initialization and parameter selection. Level set methods improve upon this by allowing contours to split, merge, and change topology automatically, making them more suitable for complex CT images.